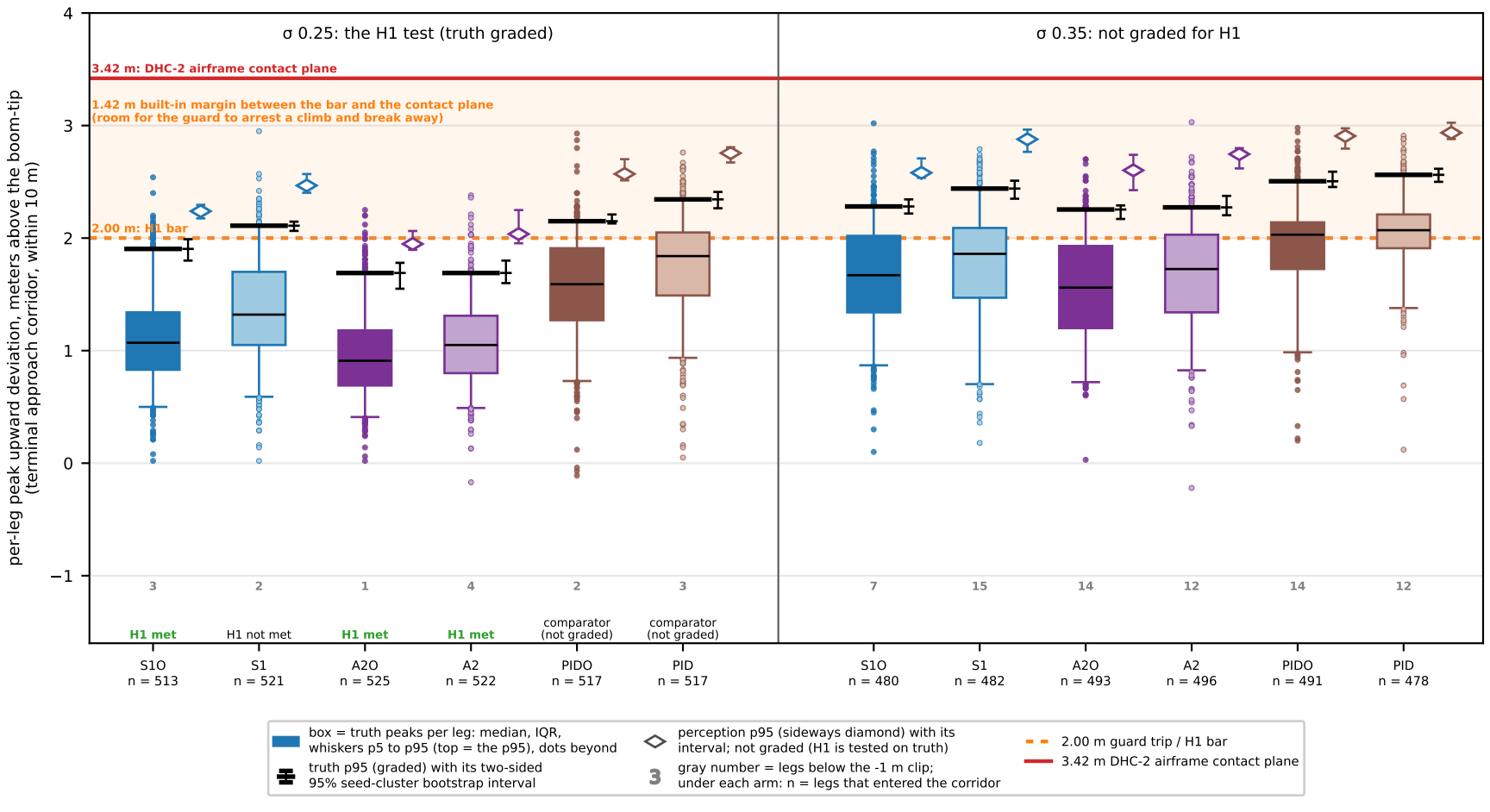


Per-Leg Peak Upward Deviation, Truth and Perception. Note. Per-leg peak upward deviation above the boom-tip inside the terminal approach corridor (within 10 m), nine wind stages pooled, 540 legs flown per arm and σ, seed block 10775-10804; n = legs that entered the corridor (the rest never came within 10 m and have no corridor peak). Boxes: truth, median, and IQR; whiskers p5 to p95, so the top whisker is the H1 statistic; dots = legs beyond the whiskers; the axis is clipped at -1 m, and legs below are counted in gray. The thick black line is the truth p95 with its two-sided 95% seed-cluster bootstrap interval; the sideways diamond is the perception p95 (what the controller read, same legs) with its interval, not graded. Left group: σ = 0.25, the H1 test; H1 is met when the truth interval lies below 2.00 m (verdict under each box). Right group: σ = 0.35, not graded for H1. Dotted orange = 2.00 m guard trip and H1 bar; solid red = 3.42 m airframe contact plane; the shaded band between them is the 1.42 m built-in margin, the room the guard has to arrest a climb and break away. Dark shades are the arms with the 0.25 m offset (S1O, A2O, PIDO); light shades the same controllers without it.